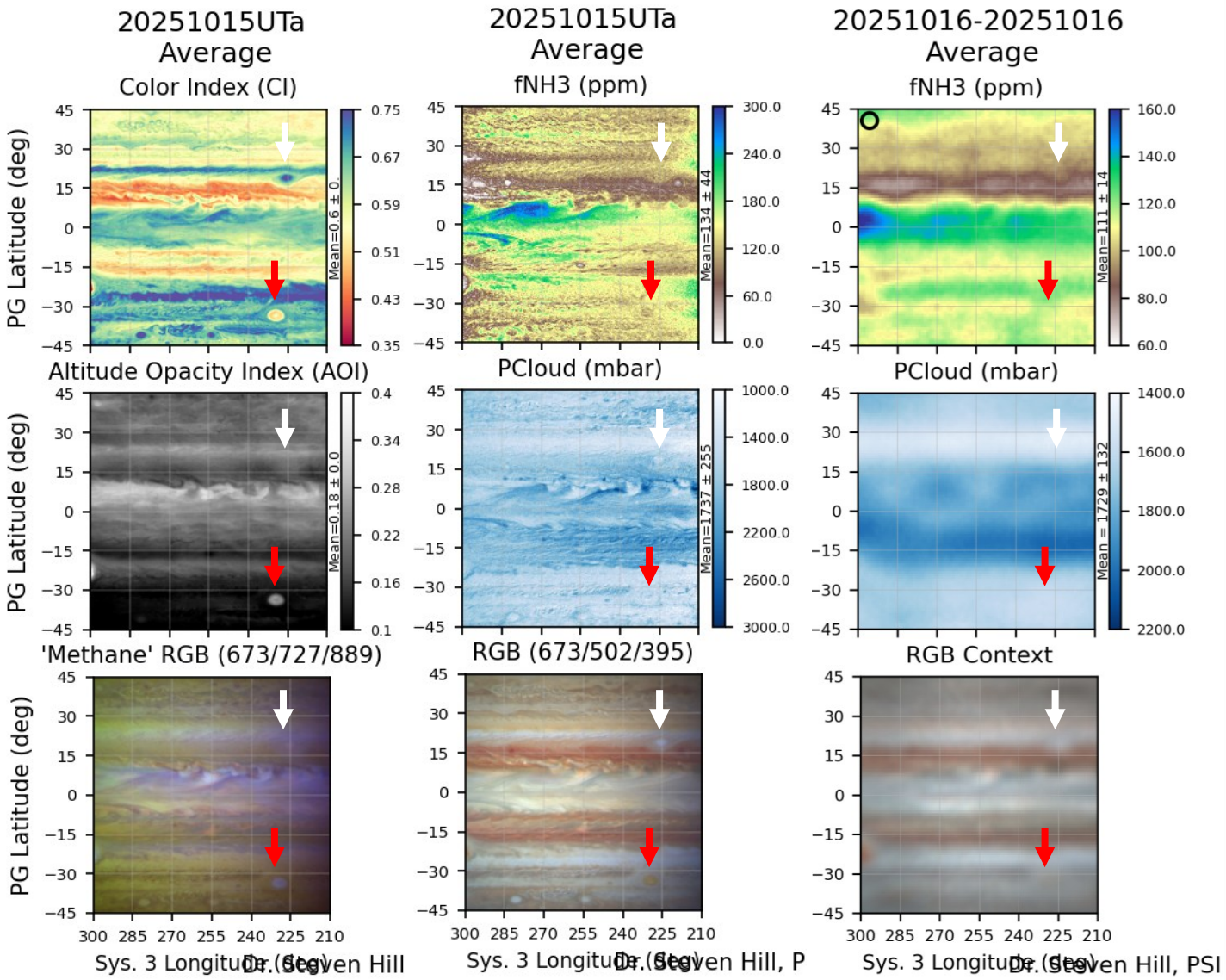
Initial Results (Oct. 15-16, 2025)

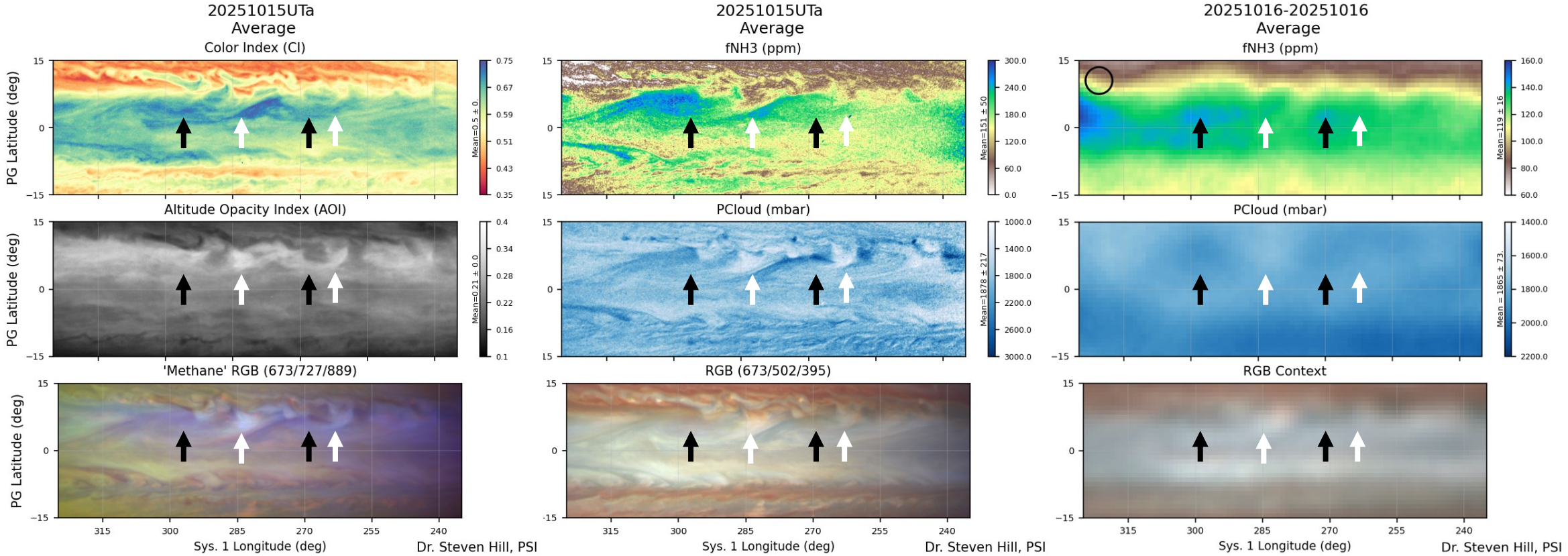
- Maps have been processed into products (for Nov. 2025 also)
  - Navigation appears to be very good
  - November data have a ~16-minute gap between the 619 nm exposure and the 631 and 645 nm exposures, which makes viewing angle corrections more challenging
  - Viewing angle correction (flattening) is done at the product-level (not radiance-level) using a 2nd order polynomial fit to large-scale longitude variation
  - Absorption coefficients: K_eff_NH3_645 = 3.090 and K_eff_CH4_619 = 0.548
- Products: AOI, CI, f_{NH3}, and P_{Cld} along with two RGB composites
  - AOI=$(I/F)_{889}/(I/F)_{275}$ and CI= $(I/F)_{395}/(I/F)_{631}$
  - f_{NH3} and P_{Cld} per Hill *et al.* (2024) and Irwin *et al.* (2025)
  - RGB(visual)=(673,502,395), RGB(NIR)=(673,727,889)

HST Data (Sys 3) from October

- EZ shows greatest variation in P_{Cld} and f_{NH3}, but neither AOI nor CI appears to correlate broadly with P_{Cld}
- White oval at 10°N (white arrow) shows slightly enriched f_{NH3} and lower P_{Cld} but barely noticeable in AOI
- AOI shows NTrZ-NTB region (17°-31°N) is multi-hued, but P_{Cld} and f_{NH3} variations are far less dramatic and poorly correlated
- Red oval at 33°S (red arrow) is prominent in AOI, but not in P_{Cld}, and shows slight f_{NH3} depletion

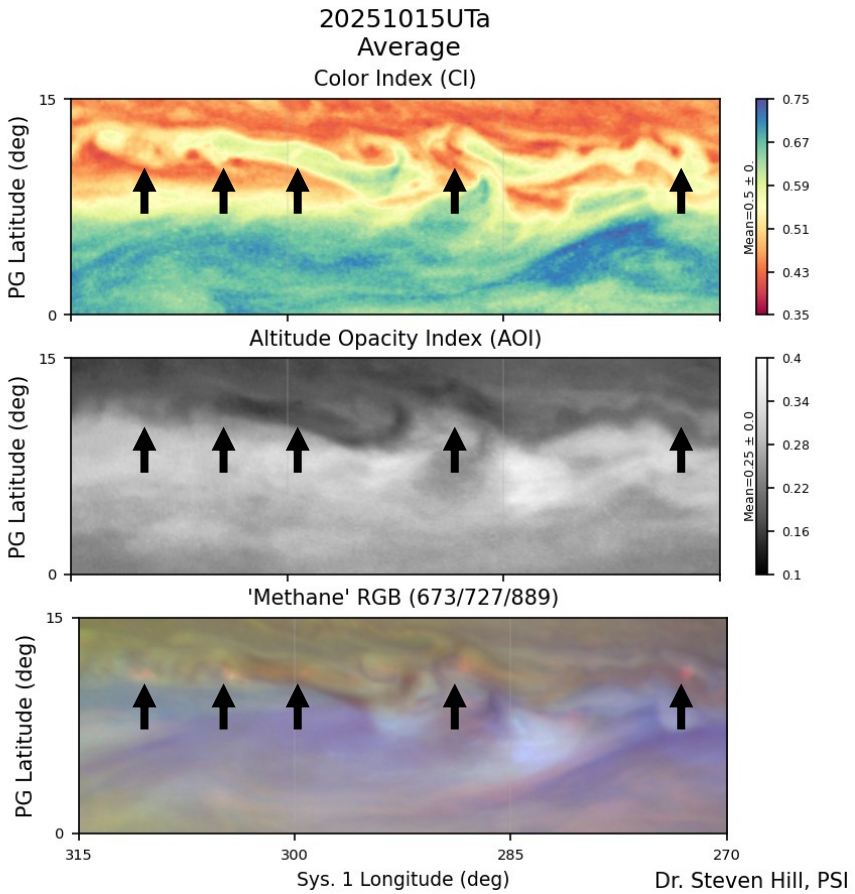


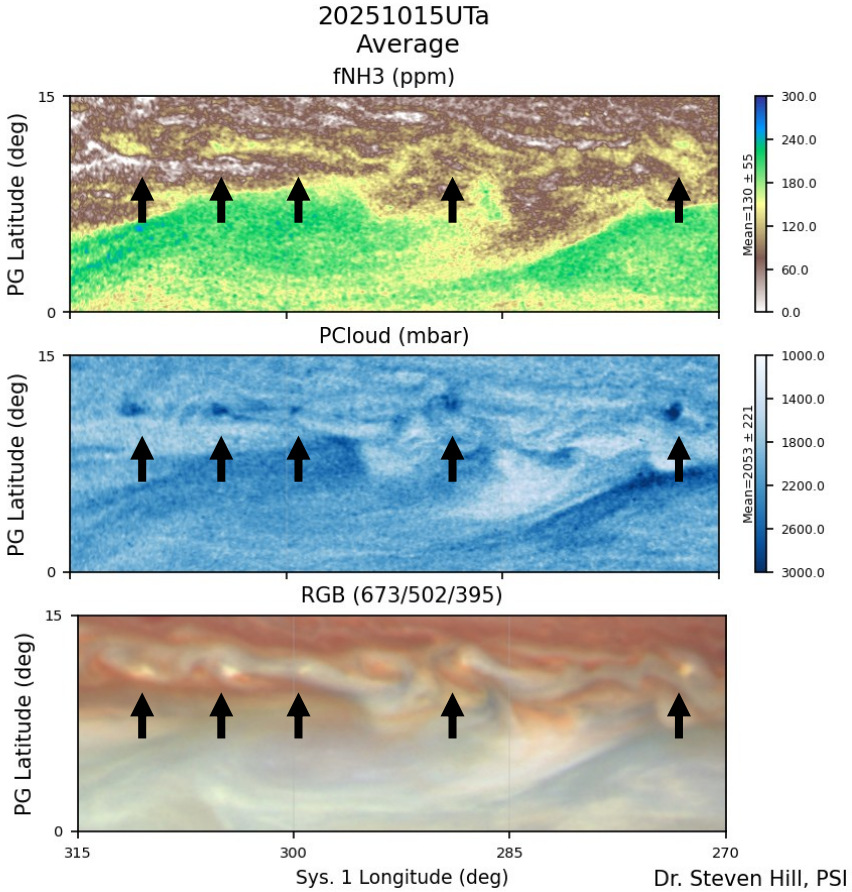
HST Data (Sys 1) from October



- High altitude bow-shaped (outflow?) clouds are seen in AOI and especially methane RGB (white arrows) also show in P_{Cld} despite the fact it 'should' be probing ~2 bar
- Hotspots (black arrows) have a complex relationship with f_{NH3}

HST Data (Sys 1) from October





- Deep clouds (red in the 'Methane RGB', lower left) show up clearly in the P_{Cld} data as dark spots (high pressure)
- No clear pattern is seen regarding f_{NH3} for these deep clouds

Summary and Continuing Work

- General Observations
  - HST f_{NH3} and P_{Cld} show greater extremes than ground-based due to higher resolution
  - Below are values from Sys. 3 map on chart 2

	HST	Ground-based
f_{NH3}	134±44 ppm	111±14 ppm
P_{Cld}	1737±255 mb	1729±132

- Continuing Work
  - Need to process December HST data
  - Need to improve flattening/limb correction, preferably to a more physics-based method, e.g., Minnaert, etc.
  - Need to produce blended global maps for each date
  - Need to produce and share products